How Long Should the Audit Run?

G1 Docket deploys in **AUDIT** first: it attests the estate, evaluates every request your AI makes, and logs what it *would* deny — while blocking nothing. The flip to **ENFORCE** is the moment of commitment, so the audit period is the one number every deployment plan has to get right. It is usually guessed. It can instead be determined — from the figures measured by the G1 Docket benchmark suite.

MID-SIZE	LARGE ENTERPRISE	HYPERSCALE	MEGA / SOVEREIGN
30 days	**60 days**	**90 days**	**90–180 days**
100K – 1M objects	1M – 10M objects	10M – 100M objects	100M – 1B objects

Every figure is **coverage-gated, not calendar-gated**: the run ends when the exit criteria on page 2 are met. The day counts are what the numbers below predict it takes to meet them.

Compute doesn't set the period. Observation does.

Start with what the scan itself costs. At the measured attestation scan rate (record evaluation across the in-memory snapshot), the entire estate is attested on day one — at every scale Docket is deployed at:

Estate (attested objects)	First full sweep	Snapshot load & verify	In-memory footprint	Full re-sweeps / day
100K (mid-market)	6.3 seconds	95 ms	~35 MB	13,700
1M (enterprise)	63 seconds	850 ms	~360 MB	1,370
10M (hyperscale)	10.5 minutes	7.8 sec	~3.6 GB	137
100M (multi-cloud)	~105 minutes	~80 sec	~35 GB	~14
1B (sovereign)	~17.5 hours	~13 min	~350 GB	delta-driven

Change never catches up either: an estate churning 5% of its objects daily re-attests in 0.3 seconds at mid-market scale, 31 seconds at ten million objects, about five minutes at a hundred million, and under an hour at a billion. The scanner is never the long pole. What takes time is **watching how the data actually moves** — and that is a property of your firm, not of the machine. (Above ten million objects the figures are linear extrapolations of the measured per-record rates; the snapshot is sharded across prevalidator nodes, each holding its segment, and re-verification runs on deltas rather than full sweeps.)

What counts as an object: a governed data object in scope for AI use — a document, a record, a dataset, a source file in a codebase Docket papers. Not every file on every disk: operating systems, dependency trees, and caches are not attested — the lockfile that names the dependencies is.

What actually sets the period

1. The window must contain the firm's periodic data. Month-end closes, payroll runs, board packs, the report someone builds once a month from a restricted spreadsheet — the most sensitive data in most firms

is also the least frequently touched. Any audit window shorter than 30 days can miss month-end entirely, depending on the day it starts. Thirty days is the floor that guarantees one full monthly cycle is inside the window.

2. Coverage follows a long tail. Enforcement is only safe when the data your AI actually uses is attested. Most of it appears in real traffic within days; the tail — quarterly artifacts, annual filings — arrives over weeks. The correct end-point is therefore a coverage threshold: at least 99% of the objects observed in live AI traffic carrying valid, unexpired attestations.

3. The would-deny log has to burn down — and prove it stayed down. AUDIT's real output is the denial forecast: every request ENFORCE would have refused, with its reason (unattested, expired, insufficient clearance, egress). Each one gets triaged — attest the data, correct a tier, or accept the denial. A mid-size estate works its log down inside one cycle; a large estate, its business units closing on staggered schedules, needs a second cycle to prove the first cycle's fixes held.

The determination

Firm scale	Estate (attested objects)	Recommended AUDIT period	The window must contain
Mid-size firm	100,000 – 1,000,000	**30 days**	One full month-end close
Large enterprise	1,000,000 – 10,000,000	**60 days**	Two month-end closes on live production traffic
Hyperscale / multi-cloud	10,000,000 – 100,000,000	**90 days**	Full quarter-end close (Q-end) across distributed BUs
Global mega-scale / sovereign data	100,000,000 – 1,000,000,000	**90 – 180 days**	Full quarter-end + semi-annual / annual reporting cycle

Why the periods lengthen: nothing about the machine changes — even a billion-object sweep is a 17.5-hour event, inside a single day. What lengthens is the calendar of the firm itself: the larger the estate, the slower its reporting heartbeat, and the window has to contain the slowest cycle that moves sensitive data. Mega-scale and sovereign estates answer to semi-annual and annual cycles too; a window that closes before those cycles fire has not watched the estate — only a sample of it.

Exit criteria — when to flip to ENFORCE

- ✓ **Coverage:** ≥ 99% of objects observed in live AI traffic hold valid, unexpired attestation records.
- ✓ **Stability:** the daily would-deny rate has been flat for seven consecutive days — no new categories of denial appearing.
- ✓ **Triage complete:** every logged would-deny has a disposition — data attested, tier corrected, or denial accepted as correct.

Meet the criteria early and you flip early — the day count is the expectation, not the sentence. What the audit period buys is certainty: on enforcement day, every denial is one you have already seen, reviewed, and accepted. There are no surprises left to have.

During the run, everything is scanned — the full estate, every change within a day of it happening, every request your AI makes — each evaluated in memory at 159 µs on average (p99 1.6 ms, inside a 5 ms budget) and logged either way. AUDIT costs the firm nothing in disruption. It costs the data nowhere to hide.

Figures are from the G1 Docket benchmark suite (in-memory snapshot model; attestation scan rate = record evaluation across the snapshot). G1 Docket is open source — the engine first, the prevalidator in front of the gateway you already run.